



POWER SOLUTIONS

Operations Selection Guide

Site Assessment & Solution Selection for Field Deployment

Document: WC-PS-OSG-v1.1 | February 2026

INTERNAL USE - OPERATIONS DEPARTMENT

Target: Project Engineers | Site Engineers | Operations Managers

TABLE OF CONTENTS

1. Quick Selection Guide 3

2. Load Calculation Fundamentals 4

3. Product Specifications Comparison 6

4. Site Assessment Checklist 8

5. Environmental Considerations 10

6. Installation Requirements 11

7. Solution Selection Decision Tree 13

8. Common Deployment Scenarios 14

1. Quick Selection Guide

Use this section for rapid solution selection based on your equipment load requirements.

1.1 Selection by Load

Total System Load	Recommended Solution	Typical Equipment	Backup Time*
≤ 8W	WakeCap Power 30 (WP-30)	WakeCap Gateway only	48+ hours
9W - 15W	WakeCap Power 60 (WP-60)	Single bullet camera, small sensor	48+ hours
16W - 100W	WakeCap Power 400 (WP-400T)	PTZ camera, multiple bullets, gateway + cameras	48+ hours
101W - 350W	WakeCap Power 900 (WP-900T)	Multiple PTZ cameras, comm equipment	48+ hours

* Backup time assumes zero solar input (worst case). Actual runtime varies with solar conditions.

CRITICAL

Always calculate TOTAL SYSTEM POWER, not just device power. System power includes the device, PoE injectors, routers, voltage converters, and all auxiliary equipment.

1.2 Equipment Power Reference

Standard power consumption for common equipment:

Equipment Type	Device Power	System Power*	Notes
WakeCap Gateway	~5W	~8W	Includes power conversion losses
Bullet Camera (standard)	15W	35W	Includes PoE injector, router
PTZ Camera (Type 1)	42W	70W	Mid-range PTZ with PoE
PTZ Camera (Type 2)	62W	90W	High-end PTZ with heater
Network Router/Switch	5-15W	—	Add to system total
PoE Injector	3-8W	—	Conversion losses

* System Power = Device + PoE + Router + Conversion Losses. Always use System Power for selection.

2. Load Calculation Fundamentals

Proper load calculation is essential for selecting the correct power solution. Follow this systematic approach.

2.1 Standard Design Parameters

All WakeCap Power Solutions are designed using these parameters:

Parameter	Symbol	Value	Description
Backup Time	T	48 hours	Required autonomy without solar input
Depth of Discharge	DoD	70%	Usable battery capacity
System Efficiency	η_{sys}	90%	Wiring and general losses
Converter Efficiency	η_{conv}	80%	DC-DC conversion losses
Solar Derating	η_{solar}	75%	Dust, misalignment, temperature
MPPT Efficiency	η_{mppt}	95%	Charge controller efficiency
Peak Sun Hours	PSH	6.5 hours	Saudi Arabia average

2.2 Load Capacity Summary

Maximum continuous loads per solution (for 48-hour backup):

Solution	Battery Capacity	Max Load (48h backup)	Max Load (24h backup)
WP-30	30Ah / 384Wh	~8W	~16W
WP-60	60Ah / 768Wh	~15W	~30W
WP-400T	400Ah / 4,800Wh	~100W	~200W
WP-900T	900Ah @ 24V / 21,600Wh	~350W	~700W

3. Product Specifications Comparison

3.1 Power System Specifications

Specification	WP-30	WP-60	WP-400T	WP-900T
System Voltage	12V DC	12V DC	12V DC	24V DC
Battery Capacity	30Ah	60Ah	400Ah (2×200)	900Ah (6×150)
Battery Energy	384Wh	768Wh	4,800Wh	21,600Wh
Battery Type	LiFePO4	LiFePO4	GEL Lead Acid	HT Lead Acid
Solar Power	80W	100W	450W	1,305W
Solar Panels	2 × 40W	2 × 50W	3 × 150W	3 × 435W
MPPT Controller	Integrated	10A	30A	60A EPEVER
MPPT Efficiency	>95%	96.5%	95%	98%
Output Voltage	12V ±0.5V	10.5-12.6V	12V DC	24V DC
Max Output Current	3A	10A	<30A (25A typical)	<60A (55A typical)

3.2 Physical Specifications

Specification	WP-30	WP-60	WP-400T	WP-900T
Total Weight	~6 kg	23 kg	525 kg	1,300 kg
Form Factor	Pole/Wall Mount	Bracket Mount	Trailer	Trailer
Mast Height	N/A	N/A	6.5m (pneumatic)	9m (manual)
Transport Size (mm)	780×530×200	940×780×240	2000×1400×2300	2320×1495×2350
Deployed Size (mm)	780×530×varies	940×780×varies	2000×2250×6500	2320×1495×2350 (+ mast)
Tow Hitch	N/A	N/A	Ball hitch	50mm ball hitch
Tire Size	N/A	N/A	165/70R13	15"

3.3 Environmental Ratings

Specification	WP-30	WP-60	WP-400T	WP-900T
IP Rating	IP66	IP66	IP65	IP65
Operating Temp	-20°C to +50°C	-30°C to +85°C	-20°C to +80°C	-20°C to +85°C
Charge Temp (Low)	-10°C	-30°C	[TBD]	[TBD]
Discharge Temp (Low)	-20°C	-30°C	[TBD]	[TBD]
Wind Rating (Panel)	2400Pa	60m/s (200kg/m²)	100 km/h	100 km/h

4. Site Assessment Checklist

Complete this assessment before selecting and deploying a Power Solution.

4.1 Load Requirements

☐ List all equipment to be powered:

- Equipment 1: _____ Power: _____ W
- Equipment 2: _____ Power: _____ W
- Equipment 3: _____ Power: _____ W
- PoE Injectors / Converters: _____ W
- Network Equipment: _____ W

☐ TOTAL SYSTEM LOAD: _____ W

☐ Required backup time without solar: _____ hours (standard: 48h)

6. Installation Requirements

6.1 Pre-Installation Checklist

- ☐ Site assessment completed (Section 4)
- ☐ Equipment manifest verified against delivery
- ☐ All components present and undamaged
- ☐ Battery state of charge verified (>40% recommended)
- ☐ Tools and PPE available
- ☐ Weather conditions acceptable (no rain, wind < 30 km/h)
- ☐ Communication with site supervisor confirmed

6.2 Solar Panel Orientation

For optimal energy harvest in Saudi Arabia:

- Panel orientation: SOUTH (0° azimuth)
- Tilt angle: 20-30° from horizontal (WP-60 adjustable 0-30°)
- Seasonal adjustment: 15° summer, 35° winter (if accessible)
- Avoid any shading, especially partial shading across cells

7. Solution Selection Decision Tree

Follow this flowchart to select the appropriate Power Solution:

[IMAGE: Decision tree flowchart: Start → Calculate Load → Select by load range → Verify site access → Confirm selection]

7.1 Decision Steps

1. Calculate TOTAL SYSTEM LOAD (not device power alone)
2. Check load against solution capacity table (Section 1.1)
3. If multiple solutions qualify, prefer smaller for cost savings
4. If borderline, select larger solution for margin

8. Common Deployment Scenarios

8.1 Scenario A: WakeCap Gateway

Situation: WakeCap Gateway for worker tracking at isolated site

- Equipment: WakeCap Gateway (~8W system)
- Solution: WP-30
- Mounting: Existing pole or temporary pole

8.2 Scenario B: Single Bullet Camera

Situation: Security camera monitoring equipment yard

- Equipment: Bullet camera + PoE + router (~35W system)
- Solution: WP-60
- Mounting: Bracket on structure or dedicated pole

8.3 Scenario C: Single PTZ Camera

Situation: PTZ surveillance camera at remote location

- Equipment: PTZ camera Type 1 (~70W system)
- Solution: WP-400T (100W capacity with margin)
- Mounting: Trailer mast (6.5m height)

8.4 Scenario D: Multi-Camera Surveillance

Situation: Construction site perimeter monitoring

- Equipment: 2× PTZ cameras + 1× bullet + P2P/Router (~200W system)
- Solution: WP-900T (350W capacity)
- Mounting: Trailer mast (9m height)
- Installation: 2-3 technicians, ~3.25 hours
- Note: Consider communication equipment power if cellular backhaul needed

8.5 Scenario E: High-Power PTZ System

Situation: High-end PTZ (with IR) with basic components in harsh environment

- Equipment: PTZ Camera (~90W) + P2P/Router (~35W) = 125W system
- Solution: WP-900T
- Note: WP-400T insufficient for 48h backup at 125W

END OF DOCUMENT

WakeCap Power Solutions - Operations Selection Guide v1.1
For further information contact: [Zishan Shahzad – zishan.shahzad@wakecap.com]